

Memory-Centric Agentic Inference — cycles 32-34

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Memory-Centric Agentic Inference — cycles 32-34

Abstract

Cycles 32-34 extended the production-evidence path for the memory-centric agentic inference architecture. The prior production chain already had validated gates for telemetry shape, adapter portability, intake custody, attestation, trust policy, deployment-root enrollment, and end-to-end gatechain replay. These cycles added two more preconditions before any real production telemetry could support DC-001/DC-002 energy and contention claims: timebase/observer-overhead integrity in cycle 32 and privacy-safe redaction with replay-identifiable joins in cycle 33.

Cycle 32 produced M-TIMEBASE-1, a timing and observer-overhead harness. It treats malformed timing as `measurement_invalid`, not as a failed threshold result. The auditor found and fixed one critical fail-closed defect: unjoinable source fixtures, identifier mismatches, nonpositive sampling periods, and negative numeric timing values could previously pass or crash. After the patch, exactly one fixture remains `timing_admissible=true`, and all rows keep `production_calibrated=false`, `production_ready=false`, and `claim_credit_allowed=false`.

Cycle 33 produced M-REDACT-1, a telemetry minimization and redaction integrity harness. It separates privacy leakage from replay non-identifiability. Under-redacted exports fail as `privacy_leakage`; over-redacted or malformed exports fail as `replay_nonidentifiable`. The auditor found and fixed one critical fail-closed defect: unknown or blank source fixtures, unsupported evidence labels, and malformed join-collision counts could previously pass or crash. After the patch, exactly one fixture remains `redaction_admissible=true`, and all rows keep `production claim credit blocked`.

Cycle 34 has supplied researcher, worker, and auditor session IDs, but no corresponding artifact family, plan entry, ledger event, or report file was present in the available workspace record. This report therefore treats cycle 34 as a reporting boundary and source-record gap, not as an additional technical milestone.

Introduction

The long-exposure run investigates whether future AI infrastructure for agentic large language model inference should be organized around memory movement, placement, reuse, compression, and lifetime management rather than arithmetic throughput alone. The broader package builds on public context for accelerators, interconnects, storage, inference benchmarks, prefix caching, semantic caching, CPU/DRAM platforms, and CXL memory expansion [1]-[14]. The local research package then adds synthetic models, executable harnesses, production telemetry contracts, and fail-closed evidence gates.

Before cycles 32-34, the project had already reached a conservative production-evidence stance. Option A, the conventional request/model/KV-serving architecture, remains the default for controls, zero-reuse workloads, cheap recomputation, high coordination overhead, or unjoinable evidence. Option B, a memory-object-aware runtime, and Option C, a trajectory/DAG-aware memory fabric, remain contract-ready pathways only when production evidence proves safe reuse or trajectory value under joined byte, power, latency, tenant, topology, security, provenance, retention, verifier, interval, and noise-floor gates.

Cycles 32 and 33 did not attempt to create production evidence. They hardened the evidence path that future production telemetry must pass. Cycle 32 asked whether telemetry intervals are temporally interpretable. Cycle 33 asked whether privacy-preserving exports still preserve the joins needed to replay memory-centric claims.

Methodology

The report consolidates completed work rather than re-auditing it. Source material came from the supplied cycle session IDs, the supplied audit report for cycle 33, `promise_ledger.jsonl`, `plan_of_record.md`, milestone markdown files, generated CSV outputs, generated figures, `REFERENCES.md`, and the current workspace manifest.

No callable session-search or session-catalog tool was available in this environment. As a result, the report uses the supplied session IDs as traceability anchors and relies on workspace artifacts, ledger entries, and the supplied audit report for technical content.

The relevant session IDs are:

Cycle	Researcher	Worker	Auditor
32	eff451cf-deac-4a2a-8043-4793a9a7efd9	8ba98cb5-9de9-4abd-a110-bcc817ca04cc	4edf2b1f-9967-46c0-b77e-8b0b1a68d817
33	b2d78ea1-957d-4b5d-89bb-de304ddd2273	adf7d656-c628-4a70-b2dc-a4d074c0777e	50fc8e79-6669-4cb2-8783-fd0a7dcaa55e
34	5b94e0fa-df9f-4d35-9232-3f10c7245f2e	8688fc50-c359-4f50-ae44-062c032bf41e	9073545a-f6ff-4b11-88b2-17e32e36afb8

Results

Context Before Cycle 32

The immediate precondition for cycle 32 was the production-evidence gatechain created in cycles 29-31. M-GATECHAIN-1 required a linear fail-closed path from raw bundle observation through attestation, trust policy, intake, adapter conformance, ingestion, security/provenance, noise-floor, threshold replay, planner update, final readiness, handoff traceability, and only then production claim credit.

Cycle 31 added M-ROOTINT-1, deployment-root enrollment and collector-root preflight replay. That gate makes root, collector, firmware, topology, schema, counter, tenant, and security bindings explicit before evidence enters the gatechain. It remains precondition-only: enrollment admissibility does not grant `production_target`, production calibration, production readiness, or claim credit.

Cycle 32 therefore started from a stricter question: even if telemetry is structurally admissible and rooted in the right deployment identity, can its timing be trusted enough to replay DC-001/DC-002 thresholds?

Cycle 32: Timebase and Observer-Overhead Integrity

Cycle 32 produced M-TIMEBASE-1, documented in `memory-centric-agent/timebase_integrity.md`. Its core result is that timing validity is a measurement-quality precondition. Invalid timing is not treated as “the threshold did not fire.” It is treated as `measurement_invalid`, which blocks threshold replay.

The harness requires a joined telemetry row to preserve known source fixture identity, measurement-run ID, bundle ID, collector ID, schema ID, clock domain, aligned power/byte/latency/queue/security intervals, nonnegative and bounded skew, nonnegative and bounded jitter, observer overhead strictly below the perturbation budget, fresh workload labels, continuous counters, and bounded drift.

The generated artifact counts after auditor validation were:

Artifact	Rows	Fields
<code>data/timebase_integrity_schema.csv</code>	36	4
<code>data/timebase_valid_fixture.csv</code>	1	40
<code>data/timebase_invalid_fixtures.csv</code>	21	40
<code>data/timebase_threshold_sensitivity_cases.csv</code>	120	11
<code>data/timebase_integrity_results.csv</code>	22	19
<code>data/timebase_failure_modes.csv</code>	21	4
<code>data/timebase_threshold_replay_boundary.csv</code>	22	7
<code>data/timebase_claim_credit_boundary.csv</code>	22	8

The auditor found one critical defect. Before the patch, the evaluator did not fully fail closed for unjoinable source fixtures, identifier-continuity mismatches, nonpositive sampling periods, and negative skew, jitter, overhead, or drift values. Some malformed rows could become timing-admissible, and some malformed numeric cases could crash rather than produce a named block.

The fix added explicit source, identifier-continuity, and numeric-domain checks. It also expanded invalid fixtures and verifier assertions. After regeneration, exactly one fixture remained `timing_admissible=true`; all malformed timing rows had named blocked reasons and `threshold_replay_status=measurement_invalid`; and every row kept `production_calibrated=false`, `production_ready=false`, and `claim_credit_allowed=false`.

The technical decision from cycle 32 is that DC-001/DC-002 replay must be timebase-aware. Missing clock domains, interval gaps, overlaps, excessive skew, stale workload labels, counter resets, and observer perturbation cannot be silently converted into threshold misses. They make the measurement uninterpretable.

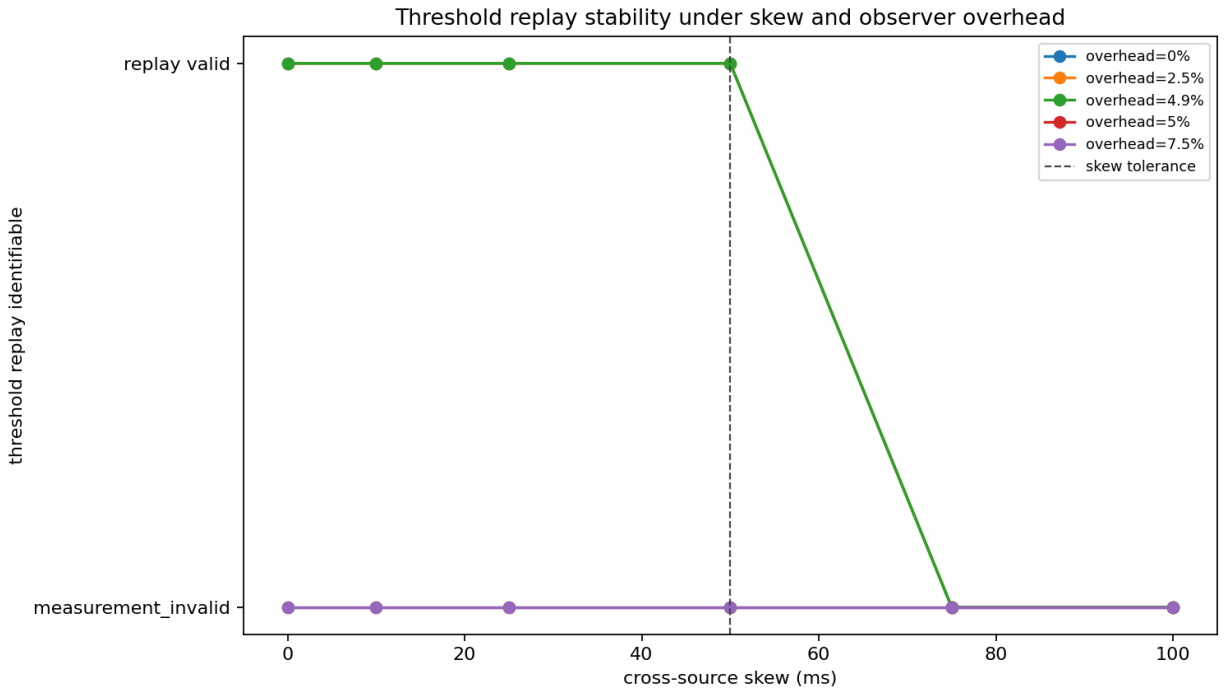


Figure 1: Threshold replay stability as cross-source skew and collector overhead increase.

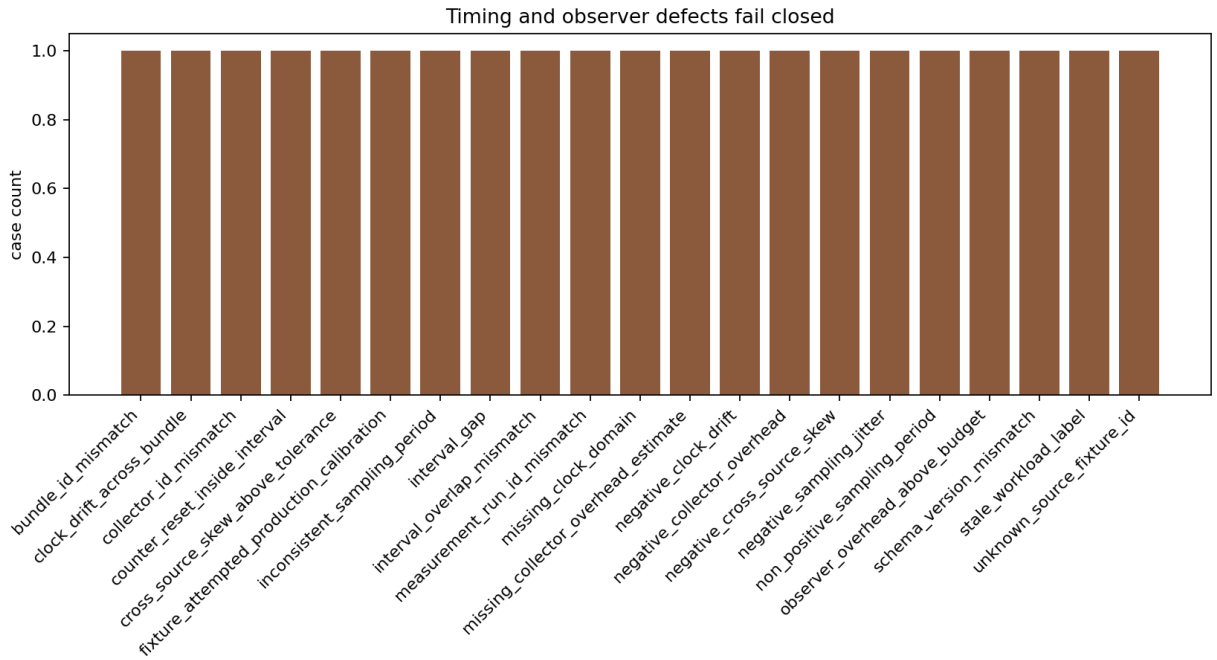


Figure 2: Fail-closed counts by timing and observer-overhead defect.

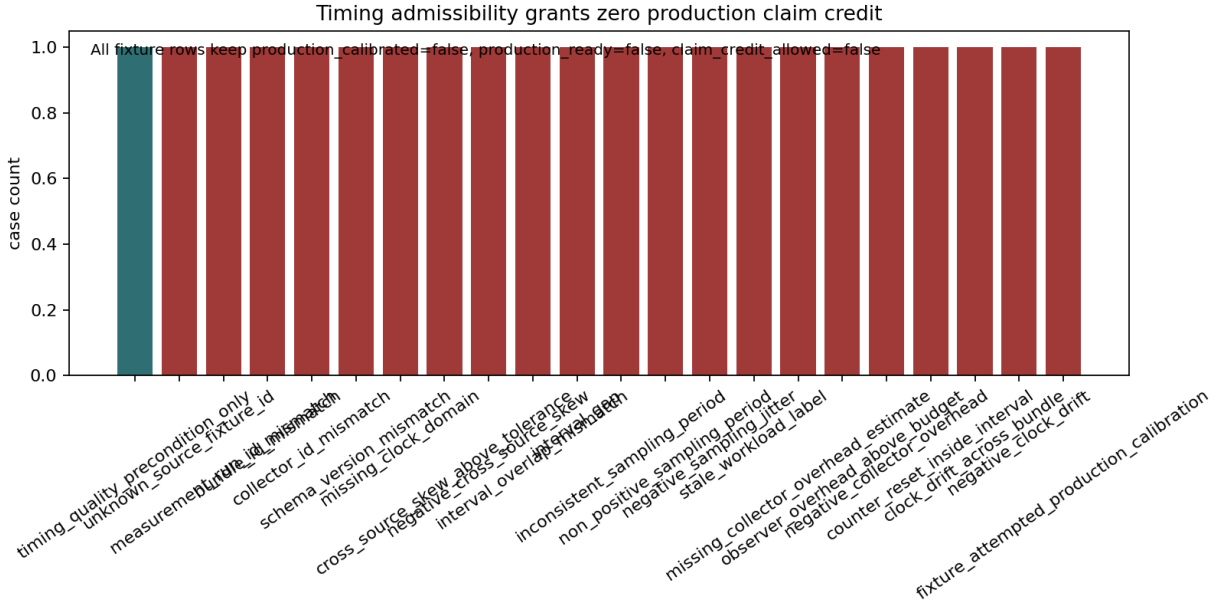


Figure 3: Timing admissibility is only a measurement-quality precondition and grants zero claim credit.

Cycle 33: Redaction and Join Preservation

Cycle 33 produced M-REDACT-1, documented in `memory-centric-agentic/redaction_integrity.md`. Its core result is that privacy-safe telemetry export must still preserve the joins required for replay. A redacted export can be admissible only if it avoids raw identifier leakage and remains identifiable enough to join back to threshold, readiness, and handoff evidence.

The redaction harness distinguishes two fail-closed outcomes:

- `privacy_leakage`: under-redacted rows expose raw tenant identifiers or raw tool-output URIs.
- `replay_nonidentifiable`: over-redacted or malformed rows destroy the joins needed to interpret DC-001/DC-002 replay.

The required join fields include stable tenant, object, run, bundle, collector, security-context, and clock-domain pseudonyms, plus workload, topology, and measurement-noise metadata at replay-compatible granularity.

The generated artifact counts after auditor validation were:

Artifact	Rows	Fields
data/redaction_integrity_schema.csv	23	4
data/redaction_policy_profiles.csv	7	5
data/redaction_valid_fixture.csv	1	23
data/redaction_invalid_fixtures.csv	20	23
data/redaction_required_join_fields.csv	11	4
data/redaction_integrity_results.csv	21	17
data/redaction_failure_modes.csv	16	4
data/redaction_join_replay_boundary.csv	21	7
data/redaction_claim_credit_boundary.csv	21	8

The auditor found one critical defect. Before the patch, unknown or blank `source_fixture_id`, unsupported or blank `evidence_label`, and negative `join_key_collision_count` values could be accepted as `redaction_admissible=true`. Nonnumeric collision counts raised `ValueError` instead of producing a named fail-closed block.

The fix added explicit source-fixture, evidence-label, and join-collision count domain checks. It also added invalid fixtures and verifier assertions for unknown or missing source fixtures, unsupported or missing evidence labels, negative collision counts, and nonnumeric collision counts. After regeneration, exactly one fixture remained `redaction_admissible=true`; under-redacted rows failed as `privacy_leakage`; over-redacted and malformed rows failed as `replay_nonidentifiable`; and every row kept `production_calibrated=false`, `production_ready=false`, and `claim_credit_allowed=false`.

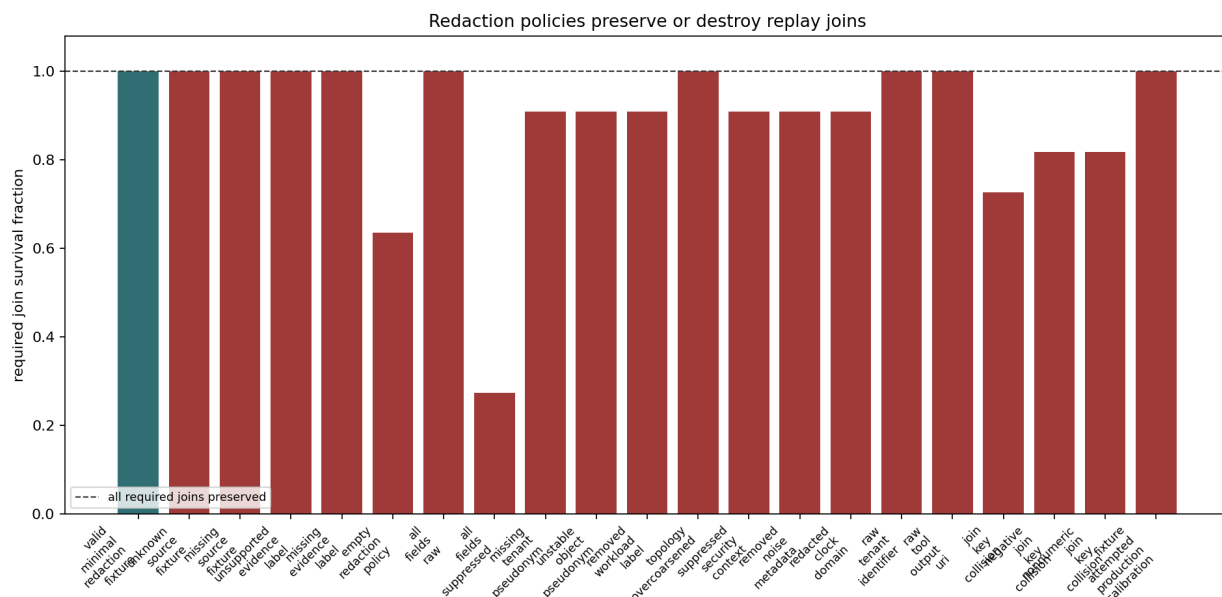


Figure 4: Fraction of required replay joins preserved by each redaction policy profile.

The technical decision from cycle 33 is that redaction cannot be evaluated only as a privacy operation. It also has to be evaluated as a replay-identifiability operation. If pseudonyms are unstable, topology is over-coarsened, clock domains are removed, security context is suppressed, or join collisions occur, the telemetry cannot support memory-centric claim replay even if it is privacy-safe.

Cycle 34: Reporting Boundary and Source-Record Gap

Cycle 34 supplied researcher, worker, and auditor session IDs, but the available workspace record contains no cycle-34 milestone entry, generated artifact family, markdown note, script, test, CSV, figure, or report artifact.

This report therefore records cycle 34 as a source-record gap. It does not infer a hidden technical result. The completed technical content in the cycles 32-34 reporting window is M-TIMEBASE-1 and M-REDACT-1.

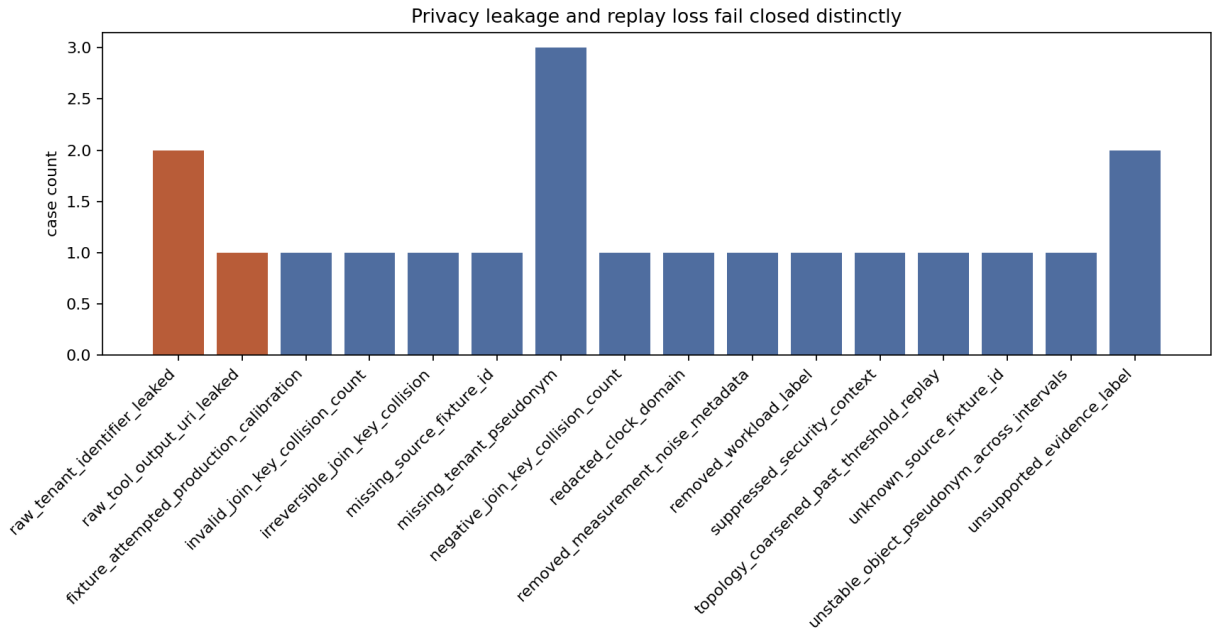


Figure 5: Fail-closed counts split between privacy leakage and replay-identifiability defects.

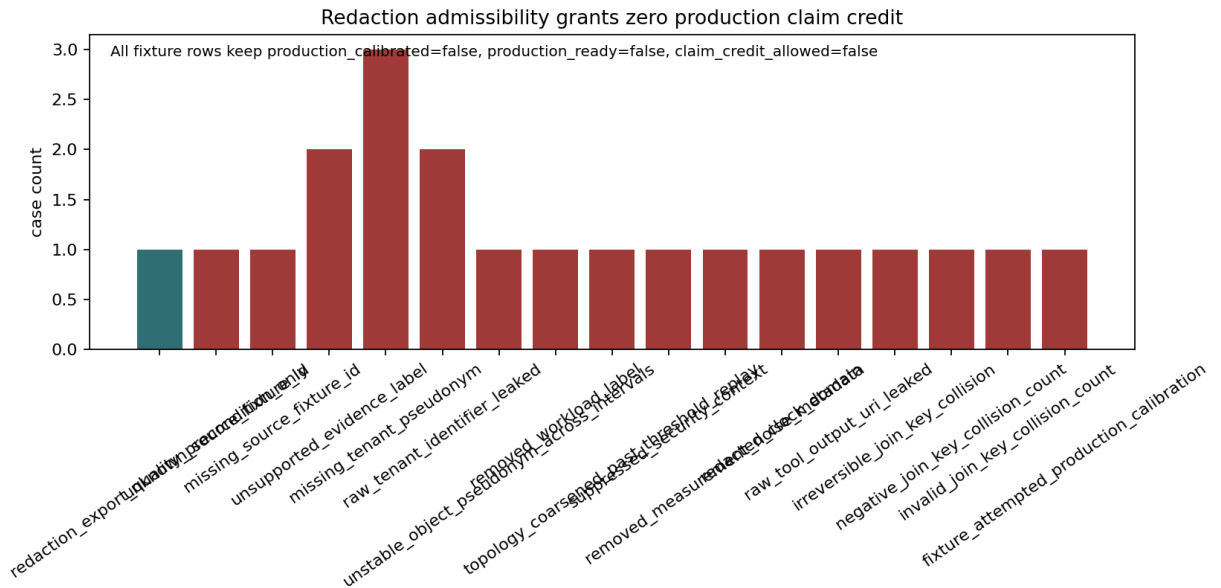


Figure 6: Redaction admissibility is an export-quality precondition and grants zero production claim credit.

Discussion

Cycles 32 and 33 narrowed the remaining path to production-ready memory-centric claims. Earlier milestones had built synthetic models, runtime prototypes, security gates, production-shaped telemetry schemas, adapter kits, custody gates, attestation envelopes, trust policies, and gatechain replay. These cycles added two more reasons why raw production-shaped telemetry is not enough.

First, threshold replay must be temporally interpretable. If measurement intervals do not align, if clocks drift outside bounds, if collector overhead perturbs the workload, or if counters reset, then DC-001/DC-002 cannot be treated as measured. The result is not a negative threshold result; it is an invalid measurement.

Second, privacy minimization must preserve replay joins. A telemetry export can be privacy-safe and still scientifically useless if it removes the tenant, object, run, bundle, collector, security, clock, workload, topology, or noise metadata needed to join evidence across gates. Conversely, preserving joins by leaking raw identifiers is not acceptable. The redaction gate forces both requirements to hold simultaneously.

The architecture stance remains unchanged. Option B and Option C remain plausible architecture options, but the project continues to block production claim credit until real `production_target` telemetry passes all gates. The current artifacts are valuable because they make the required production path more explicit and falsifiable. They do not by themselves prove production energy, latency, contention, or cost advantages.

Conclusions and Recommendations

Cycle 32 validated M-TIMEBASE-1. The timing harness ensures that malformed timebase, interval, sampling, counter, skew, jitter, drift, or observer-overhead evidence fails closed as `measurement_invalid`. Timing admissibility remains a precondition only.

Cycle 33 validated M-REDACT-1. The redaction harness ensures that privacy-unsafe exports fail as `privacy_leakage` and replay-destroying exports fail as `replay_nonidentifiable`. Redaction admissibility remains a precondition only.

Cycle 34 has no technical artifact in the available workspace record.

The next meaningful advancement remains real trusted `production_target` telemetry replayed through the full evidence path: deployment-root enrollment, attestation, intake custody, timebase validity, redaction admissibility, adapter conformance, gatechain replay, security/provenance validation, noise-floor checks, DC-001/DC-002 threshold replay, planner update, final readiness, and handoff traceability.

Further synthetic gates are likely diminishing returns unless they test a new fail-closed boundary not already covered.

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Appendix: Implementation Details

Code Organization

Cycle 32 added or used the following primary files:

File	Purpose
memory-centric-agentic/timebase_integrity.md	Narrative for timebase and observer-overhead integrity.
scripts/build_timebase_integrity_fixtures.py	Builds schema, valid fixture, invalid fixtures, and sensitivity cases.
scripts/evaluate_timebase_integrity.py	Evaluates timing admissibility and measurement-invalid outcomes.
scripts/plot_timebase_integrity.py	Renders timebase figures.
tests/verify_timebase_integrity.py	Verifies schema, invalid fixtures, sensitivity cases, boundaries, and figures.

Cycle 33 added or used the following primary files:

File	Purpose
memory-centric-agentic/redaction_integrity.md	Narrative for redaction, privacy leakage, and replay identifiability.
scripts/build_redaction_integrity_fixtures.py	Builds schema, policy profiles, valid fixture, invalid fixtures, and required join-field map.
scripts/evaluate_redaction_integrity.py	Evaluates privacy leakage, replay identifiability, redaction admissibility, and claim boundaries.
scripts/plot_redaction_integrity.py	Renders redaction figures.
tests/verify_redaction_integrity.py	Verifies redaction schema, failure modes, join preservation, boundaries, and figures.

Figure Inventory

Figure	Dimensions	Role
data/timebase_skew_sensitivity.png	1520 x 880	Shows skew and observer-overhead sensitivity for threshold replay.
data/timebase_failure_modes.png	1680 x 928	Shows fail-closed timing and observer-overhead defects.
data/timebase_claim_boundary.png	1632 x 832	Shows timing admissibility as a no-claim-credit precondition.
data/redaction_join_survival.png	1920 x 960	Shows required replay-join survival across redaction profiles.
data/redaction_failure_modes.png	1760 x 928	Shows privacy leakage and replay-identifiability failures.
data/redaction_claim_boundary.png	1680 x 832	Shows redaction admissibility as a no-claim-credit precondition.

Validation Results

For M-TIMEBASE-1, the auditor reported that the build, evaluation, plotting, verifier, Python compilation, independent fail-closed probes, CSV/figure checks, promise_check, and org_check passed. org_check retained only known root-file warnings.

For M-REDACT-1, the auditor reported that the build, evaluation, plotting, verifier, Python compilation, independent fail-closed probes, CSV/figure checks, promise_check, and org_check passed. org_check retained only known root-file warnings for CURATION.yaml and two package zip files.

Source Session References

Source	How it is used
Cycle 32 researcher eff451cf-deac-4a2a-8043-4793a9a7efd9	Traceability anchor for the cycle 32 timebase objective.
Cycle 32 worker 8ba98cb5-9de9-4abd-a110-bcc817ca04cc	Traceability anchor for M-TIMEBASE-1 implementation artifacts.
Cycle 32 auditor 4edf2b1f-9967-46c0-b77e-8b0b1a68d817	Traceability anchor for M-TIMEBASE-1 validation and critical patch.
Cycle 33 researcher b2d78ea1-957d-4b5d-89bb-de304ddd2273	Traceability anchor for the cycle 33 redaction objective.
Cycle 33 worker adf7d656-c628-4a70-b2dc-a4d074c0777e	Traceability anchor for M-REDACT-1 implementation artifacts.
Cycle 33 auditor 50fc8e79-6669-4cb2-8783-fd0a7dcaa55e	Traceability anchor for M-REDACT-1 validation and critical patch.
Cycle 34 researcher 5b94e0fa-df9f-4d35-9232-3f10c7245f2e	Supplied ID only; no corresponding workspace artifact found.
Cycle 34 worker 8688fc50-c359-4f50-ae44-062c032bf41e	Supplied ID only; no corresponding workspace artifact found.
Cycle 34 auditor 9073545a-f6ff-4b11-88b2-17e32e36afb8	Supplied ID only; no corresponding workspace artifact found.

Cross-Reference Map

Origin	Consuming artifact	Flow
data/production_root_enrollment_results.csv	data/timebase_valid_fixture.csv, data/timebase_integrity_results.csv	Deployment-root and collector-root identity continuity is upstream of timing-admissible telemetry.
data/timebase_integrity_results.csv	data/redaction_valid_fixture.csv, data/redaction_integrity_results.csv	Redaction integrity assumes a known timebase-admissible source fixture.
data/timebase_threshold_replay_boundary.csv	data/timebase_claim_credit_boundary.csv	Timing-valid threshold replay still grants no production claim credit.
data/redaction_required_join_fields.csv	data/redaction_join_replay_boundary.csv	Stable pseudonym and coarsening rules determine whether redacted telemetry remains replay-identifiable.
data/redaction_integrity_results.csv	data/redaction_claim_credit_boundary.csv	Redaction admissibility still grants no production claim credit.
data/evidence_gatechain_replay_results.csv	timebase and redaction gates	The broader gatechain remains the downstream promotion path for future production evidence.

Manifest Snapshot

MANIFEST.md was updated to reflect the cycles 32-34 workspace state. The current snapshot records 70 Python scripts in scripts/, 4 Wolfram scripts, 19 test scripts, 33 markdown

model/synthesis files under memory-centric-agentic/, 182 CSV files under data/, 83 figures under data/, and 32 completed, assessed, or designed subtopics. It adds M-ROOTINT-1, M-TIMEBASE-1, and M-REDACT-1 to the validation and cross-reference summaries.